PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

Programming Fundamentals Lab 02	**Instructor: Mr. Muhammad Hamza**
Program: AI- Fall 2025	**Date: 16th October, 2025**
Marks: 10	**Lab Engineer: Ms. Qandil Fatima**
Student Name: Eshal Ayub	**Reg. No: B25F1061AI100**
Instructions: • Submit the file with your names following your registration numbers like **AI001_Name.** • **Copy Paste work will be given no marks.**	

Lab Task

Type and save the following programs in Visual Studio. Run these programs and observe their output.

[*] Untitled1.cpp

```cpp
#include <iostream>
using namespace std;

int main()
{
char alp;
    alp = 'A';
    cout << alp << '\n';
    alp = 'B';
    cout << alp << '\n';
    cin >> alp;


    char letter;
    letter = 65;
    cout << letter << '\n';
    letter = 66;
    cout << letter << '\n';
    cin >> letter;


    int alpha;
    letter = 'A';
    cout << alpha << '\n';
    letter = 'B';
    cout << alpha << '\n';
    cin >> alpha;
}
```

Output:

C:\Users\qandi\OneDrive - Higher Educa

```
Please enter your 3 numbers
4
5
6
4
5
6
6
5
4
Press any key to continue . .
```

1. Ask users to enter 2 numbers from keyboard and apply following operations (+,*,-,/,%) on them and then print output:

Untitled1.cpp

```cpp
#include <iostream>
using namespace std;

int main()
{
int a, b;
cout<< "enter two numbers:";
cin >> a >> b;

cout << "Addition " << a + b << endl;
cout << "Subtraction " << a - b << endl;
cout << "Multiplication " << a * b << endl;
cout << "Division " << a / b << endl;
cout << "Modulus " << a % b << endl;

return 0;
}
```

Output:

C:\Users\DELL PRECision 7550

```
enter two numbers:2
3
Addition 5
Subtraction -1
Multiplication 6
Division 0
Modulus 2

--------------------------------
Process exited after 41.96 seconds with return value 0
Press any key to continue . . .
```

2. Write a C++ program to find the distance between two points P1(x1,y1) and P2(x2,y2)? Formula:

$$D = \sqrt{(x2 - x1)2} + (y2 - y1)2$$

[*] Untitled1.cpp

```cpp
#include <iostream>
#include <cmath>// allows use of math functions like sqrt(), pow(), sin(), etc.
using namespace std;

int main()
{
float x1, y1, x2, y2, distance;

cout << "Enter x1 and y1:";
    cin >> x1 >> y1;
    cout << "Enter x2 and y2: ";
    cin >> x2 >> y2;

distance = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));// using distance formula: v((x2 - x1)² + (y2 - y1)²)
cout << "Distance between two points = " << distance << endl;

return 0;
}
```

Output:

C:\Users\DELL PRECision 7550

```
Enter x1 and y1:4
3
Enter x2 and y2: 2
5
Distance between two points = 2.82843

--------------------------------
Process exited after 17.48 seconds with return value 0
Press any key to continue . . .
```

3. Compute area of Circle ($A = \pi r^2$), Sphere ($S = 4\pi r^3$) and Triangle ($A = \frac{1}{2}bh$). Ask user to enter parameters and check if the values entered are correct?

[*] Untitled1.cpp

```cpp
#include <iostream>
#include <cmath>// allows use of math functions like sqrt(), pow(), sin(), etc.
using namespace std;

int main()
{
 float r, b, h;
    const float pi = 3.14159;

    cout << "Enter radius of circle and sphere: ";
    cin >> r;
    cout << "Enter base and height of triangle: ";
    cin >> b >> h;

    float circleArea = pi * r * r;
    float sphereArea = 4 * pi * r * r;
    float triangleArea = 0.5 * b * h;

    cout << "Area of Circle = " << circleArea << endl;
    cout << "Area of Sphere = " << sphereArea << endl;
    cout << "Area of Triangle = " << triangleArea << endl;

return 0;
}
```

Output:

C:\Users\DELL PRECision 7550

```
Enter radius of circle and sphere: 3
Enter base and height of triangle: 2
6
Area of Circle = 28.2743
Area of Sphere = 113.097
Area of Triangle = 6

--------------------------------
Process exited after 24.14 seconds with return value 0
Press any key to continue . . .
```

4. A customer in a store purchases following items. The prices of the five items are:

Price of item 1 = $12.95

Price of item 2 = $24.95

Price of item 3 = $6.95

Price of item 4 = $14.95

Price of item 5 = $3.95

Write a program in C++ that holds the prices of the five items in five variables. Display each items price, the subtotal of the sale, the amount of sales tax, and the total. Assume the sales tax is 6%.

Sub_total = price1+price2+price3+price4+price5;

Sales_tax = sub_total * 0.06;

Total_bill = Subtotal + sales tax;

[*] Untitled1.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    // Step 1: Declare and assign prices of 5 items
    double price1 = 12.95;
    double price2 = 24.95;
    double price3 = 6.95;
    double price4 = 14.95;
    double price5 = 3.95;

    // Step 2: Calculate subtotal
    double subtotal = price1 + price2 + price3 + price4 + price5; //double is used to store decimal values (like money).

    // Step 3: Calculate sales tax (6% of subtotal)
    double salesTax = subtotal * 0.06;

    // Step 4: Calculate total bill
    double total = subtotal + salesTax;

    // Step 5: Display everything
    cout << "Price of item 1 = $" << price1 << endl;
    cout << "Price of item 2 = $" << price2 << endl;
    cout << "Price of item 3 = $" << price3 << endl;
    cout << "Price of item 4 = $" << price4 << endl;
    cout << "Price of item 5 = $" << price5 << endl;

    cout << "\nSubtotal = $" << subtotal << endl;
    cout << "Sales Tax (6%) = $" << salesTax << endl;
    cout << "Total Bill = $" << total << endl;

    return 0;
}
```

Output:

C:\Users\DELL PRECision 7550

```
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95

Subtotal = $63.75
Sales Tax (6%) = $3.825
Total Bill = $67.575

--------------------------------
Process exited after 0.08765 seconds with return value 0
Press any key to continue . . .
```